Project Name: Movie Recommendation System – COMPSCI 2ME3 (Team#65)

Test Stage: System Testing

Test Date: Dec 5, 2025

Test Case ID#: 1

Name of Tester: Ibrahim Awan

Test Description: Validation of REQ-1: The system must be able to read each record in the movies.txt file into a java object. These objects, when aggregated into a list, represent the movie master catalog that can be used by the recommendation system.

Results (P/F): P

Preconditions for Test: Use default movies.txt consisting of 100 records

Step #	Test Step Description	Test Data	Expected Result	Actual Result (terminal output)	Notes
1	Launch the application	Default movies.txt	On application startup, the terminal displays the number of movie records successfully parsed and translated into unique Movie java objects		

Post condition(s) for Test: N/A

Test Stage: System Testing

Test Date: Dec 5, 2025

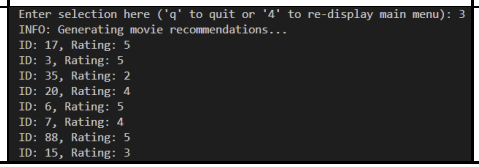
Test Case ID#: 2

Name of Tester: Ibrahim Awan

Test Description: Validation of REQ-2: The system must be able to read each record in the my_ratings.txt file into a key-value pair mapping a movie id to a user rating. These kv pairs, when aggregated, represent the movies that have been rated by the user along with their ratings.

Results (P/F): P

Preconditions for Test: Use default my_ratings.txt consisting of 8 ratings. Temporarily, add logging code at the beginning of *RecommendationEngine.generateMovieRecommendations()* method to display the output of *FileHandler.fetchUserRatings()* to the terminal

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Launch the application	Default my_ratings.txt	Normal startup	Normal startup	
2	When prompted for a selection, choose option “3”	Enter “3”	Due to the logging code, the terminal displays the 8 user ratings stored in my_ratings.txt		

Post condition(s) for Test: Remove the logging code

Test Stage: Unit Testing

Test Date: Dec 5, 2025

Test Case ID#: 3

Name of Tester: Ibrahim Awan

Test Description: Validation of REQ-3: The system must be able to gracefully handle parsing errors encountered when parsing movies.txt. If an error is encountered, the system must log the error to the console and continue with valid data. Refer to Section 3.3.3 to see the format of movie data in movies.txt.

Results (P/F): P

Preconditions for Test: N/A

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Run *validateMovieParsingLogic()* test method in the FileHandlerTest class	View the @ValueSource annotation that is used to specify the array of String arguments for the test method	Since all the string arguments represent invalid movie data, the system should handle the parsing errors and display the error to the terminal	See below	This functional requirement is better tested using a unit test rather than system testing. The *validateMovieParsingLogic()* test method in the FileHandlerTest class tests how the primary parsing method in the application handles malformed movie data. The primary parsing method is *parseMovieRecord()* in the FileHandler class.

Actual Result:

```
WARNING: Error parsing the following record: [The Shawshank Redemption,Frank Darabont,1994,Drama,4.9,2847]. Error msg: (Missing movie information). Proceeding without this data
WARNING: Error parsing the following record: [2,The Godfather,Francis Ford Coppola,Crime;Drama,4.8,2156]. Error msg: (Missing movie information). Proceeding without this data
WARNING: Error parsing the following record: [3,The Dark Knight,Christopher Nolan,200rewf8,Action;Crime;Drama,4.7,3421]. Incorrect data type used for movie data (For input string: "200rewf8"). Proceeding without this data
WARNING: Error parsing the following record: [4,Pulp Fiction,Quentin Tarantino,1994,Crime;Drama,NO_RATING,2698]. Incorrect data type used for movie data (For input string: "NO_RATING"). Proceeding without this data
WARNING: Error parsing the following record: [5,Forrest Gump,Robert Zemeckis,1994,,4.5,2134]. Error msg: (Movie genres are missing). Proceeding without this data
WARNING: Error parsing the following record: [6,Inception,Christopher Nolan,2010,NO_VALID_GENRE,4.7,3156]. Error msg: (unknown genre: no_valid_genre). Proceeding without this data
WARNING: Error parsing the following record: [7,The Matrix,The Wachowskis,1999,;;;;,4.6,2876]. Error msg: (movie genres are missing). Proceeding without this data
WARNING: Error parsing the following record: [8,Goodfellas,Martin Scorsese,1990,Biography;Crime;Drama,4.7,-1234]. Incorrect data type used for movie data (Illegal leading minus sign on unsigned string -1234.). Proceeding without this data
WARNING: Error parsing the following record: [9,The Silence of the Lambs,Jonathan Demme,1991,Crime;Drama;Thriller,-2.4234223,1876]. Error msg: (Movie average rating cannot be negative). Proceeding without this data
WARNING: Error parsing the following record: [10.234,Saving Private Ryan,Steven Spielberg,1998,Drama;War,4.7,1654]. Incorrect data type used for movie data (For input string: "10.234"). Proceeding without this data
WARNING: Error parsing the following record: [11,The Lord of the Rings: The Return of the King,Peter Jackson,2003,Adventure;Dr4m4;Fantasy,4.8,2234]. Error msg: (unknown genre: dr4m4). Proceeding without this data
```

Post condition(s) for Test: N/A

Test Stage: System Testing

Test Date: Dec 5, 2025

Test Case ID#: 4a

Name of Tester: Ibrahim Awan

Test Description: Validation of REQ-4: If movies.txt does not exist or is empty, the system must log a message to the console and exit gracefully since the system cannot function without movie data.

Test Strategy: Test with non-existent movies.txt file

Results (P/F): P

Preconditions for Test: Delete (or preferably, rename) the default movies.txt file

Step #	Test Step Description	Test Data	Expected Result	Actual Result (terminal output)	Notes
1	Launch the application	N/A	On application startup, the terminal displays an error message indicating that it cannot find the movies.txt file. The application then exits gracefully	PS C:\Users\ibrah\Desktop\COMPSCI 2ME3\project\2me3-group-project-group-65\MovieRecommendationSystem> c:; cd 'c:\Users\ibrah\Desktop\COMPSCI 2ME3\project\2me3-group-project-group-65\MovieRecommendationSystem'; & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\ibrah\Desktop\COMPSCI 2ME3\project\2me3-group-project-group-65\MovieRecommendationSystem\target\classes' 'mcmaster_compsci2me3.MovieRecommendationSystem' ERROR: movies.txt does not exist in data directory! Exiting application... PS C:\Users\ibrah\Desktop\COMPSCI 2ME3\project\2me3-group-project-group-65\MovieRecommendationSystem>	

Post condition(s) for Test: Restore movies.txt

Test Stage: System Testing

Test Date: Dec 5, 2025

Test Case ID#: 4b

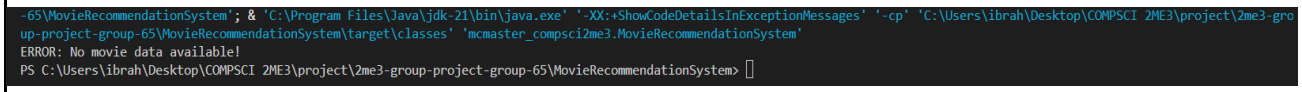
Name of Tester: Ibrahim Awan

Test Description: Validation of REQ-4: If movies.txt does not exist or is empty, the system must log a message to the console and exit gracefully since the system cannot function without movie data.

Test Strategy: Delete all movie data from movies.txt (i.e. empty movies.txt)

Results (P/F): P

Preconditions for Test: Delete all movie records from default movies.txt file

Step #	Test Step Description	Test Data	Expected Result	Actual Result (terminal output)	Notes
1	Launch the application	N/A	On application startup, the terminal displays an error message indicating that there is no movie data available. The application then exits gracefully		This error message appears regardless of whether the header file is present in movies.txt

Post condition(s) for Test: Restore movies.txt

Test Stage: System Testing

Test Date: Dec 5, 2025

Test Case ID#: 5

Name of Tester: Ibrahim Awan

Test Description: Validation of REQ-5: If my_ratings.txt does not exist, the system must treat this as normal for a new user and create the file when the user rates their first movie.

Results (P/F): P

Preconditions for Test: Delete (or preferably, rename) the default my_ratings.txt file

Step #	Test Step Description	Test Data	Expected Result	Actual Result (terminal output)	Notes
1	Launch the application	N/A	On application startup, the terminal displays a message acknowledging a cold start. A my_ratings.txt file is created in the data/ folder	INFO: Cold start detected... Created new my_ratings.txt file INFO: movies.txt successfully parsed - 100 total records found (0 records filtered out) Welcome to the Movie Recommendation System! Enter '1' to rate a movie Enter '2' to display movie catalog Enter '3' to generate movie recommendations Enter selection here ('q' to quit or '4' to re-display main menu):	

Post condition(s) for Test: Restore my_ratings.txt

Test Stage: System Testing

Test Date: Dec 5, 2025

Test Case ID#: 6

Name of Tester: Ibrahim Awan

Test Description: Validation of REQ-6: Generalizing REQ's 3-5, the system must never crash due to missing files or malformed data.

Results (P/F): N/A. This requirement is implicitly tested throughout the testing phase

Test Stage: System Testing

Test Date: Dec 5, 2025

Test Case ID#: 7

Name of Tester: Ibrahim Awan

Test Description: Validation of REQ-7: The system must allow the user to display the movie catalog in the terminal so that users can view which movies are available for them to rate. This requirement references command 2 in Section 3.1.3.

Results (P/F): P

Preconditions for Test: Use default movies.txt and my_ratings.txt files

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Launch the application	Default data files	Normal startup	Normal startup	
2	When prompted for a selection, choose option “2”	Enter “2”	The user can see which movies are available to them to rate (or re-rate)	See below	

```
Enter selection here ('q' to quit or '4' to re-display main menu): 2
{id: 1, title: The Shawshank Redemption, director: Frank Darabont, year: 1994, genres: [DRAMA]}
{id: 2, title: The Godfather, director: Francis Ford Coppola, year: 1972, genres: [CRIME, DRAMA]}
{id: 3, title: The Dark Knight, director: Christopher Nolan, year: 2008, genres: [ACTION, CRIME, DRAMA]}
{id: 4, title: Pulp Fiction, director: Quentin Tarantino, year: 1994, genres: [CRIME, DRAMA]}
{id: 5, title: Forrest Gump, director: Robert Zemeckis, year: 1994, genres: [DRAMA, ROMANCE]}
{id: 6, title: Inception, director: Christopher Nolan, year: 2010, genres: [ACTION, SCI_FI, THRILLER]}
{id: 7, title: The Matrix, director: The Wachowskis, year: 1999, genres: [ACTION, SCI_FI]}
{id: 8, title: Goodfellas, director: Martin Scorsese, year: 1990, genres: [BIOGRAPHY, CRIME, DRAMA]}
{id: 9, title: The Silence of the Lambs, director: Jonathan Demme, year: 1991, genres: [CRIME, DRAMA, THRILLER]}
{id: 10, title: Saving Private Ryan, director: Steven Spielberg, year: 1998, genres: [DRAMA, WAR]}
{id: 11, title: The Lord of the Rings: The Return of the King, director: Peter Jackson, year: 2003, genres: [ADVENTURE, DRAMA, FANTASY]}
{id: 12, title: The Lord of the Rings: The Fellowship of the Ring, director: Peter Jackson, year: 2001, genres: [ADVENTURE, DRAMA, FANTASY]}
{id: 13, title: Star Wars: Episode V - The Empire Strikes Back, director: Irvin Kershner, year: 1980, genres: [ACTION, ADVENTURE, FANTASY]}
{id: 14, title: Schindler's List, director: Steven Spielberg, year: 1993, genres: [BIOGRAPHY, DRAMA, HISTORY]}
{id: 15, title: Fight Club, director: David Fincher, year: 1999, genres: [DRAMA]}
{id: 16, title: The Lord of the Rings: The Two Towers, director: Peter Jackson, year: 2002, genres: [ADVENTURE, DRAMA, FANTASY]}
```

Post condition(s) for Test: N/A

Test Stage: System Testing

Test Date: Dec 5, 2025

Test Case ID#: 8

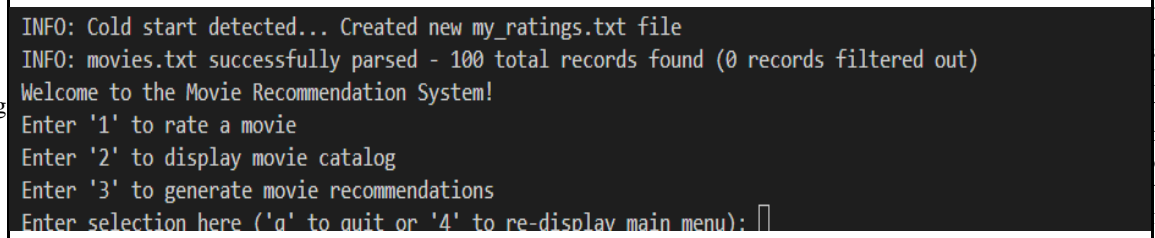
Name of Tester: Ibrahim Awan

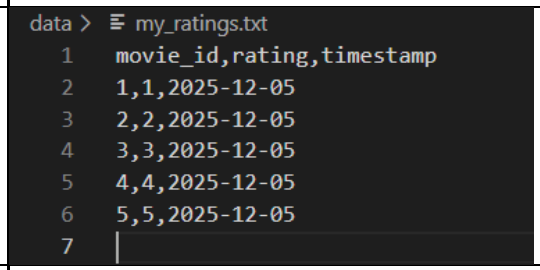
Test Description: Validation of REQ-8 and REQ-9: The system must be able to allow users to rate movies with persistence to my_ratings.txt, including the ability to update existing ratings. The system must be also able to handle a cold start (i.e. when my_ratings.txt does not exist). In this case, it should create a new file on application startup

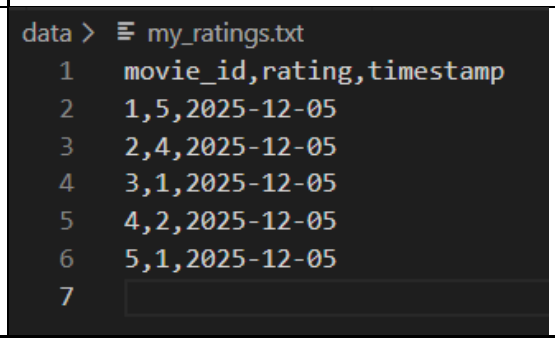
Test Strategy: Start the application without a my_ratings.txt file and show that the system can create the ratings file and write user ratings to the file with persistence. Also show that existing ratings can be overwritten by the user

Results (P/F): P

Preconditions for Test: Delete (or preferably, rename) the default my_ratings.txt file to simulate cold-start

Step #	Test Step Description	Test Data	Expected Result	Actual Result (terminal output)	Notes
1	Launch the application	Default movies.txt	On application startup, the terminal displays a message acknowledging a cold start. A my_ratings.txt file is created in the data/ folder		Realized during the process of writing these test cases that REQ-9 is a duplicate of REQ-5. If we had more time, we would have revisited the SRS document to remove the duplicated functional requirement
2	When prompted for a selection, choose option "1"	Enter "1"	System prompts the user to enter the movie id to rate	Matches expectation	Because this is a required step if the user wants to repeatedly rate movies, it is only shown once here to avoid duplication
3	When prompted	Enter "1"	System	Matches expectation	Although

	for a movie id and rating to assign, choose “1” and “1”	twice	notifies the user that the rating has been saved and a new record is written to my_ratings.txt		my_ratings.txt is updated after every operation, a screenshot of the file is only shown after step 7 for readability
4	When prompted for a movie id and rating to assign, choose “2” and “2”	Enter “2” twice	“”	“”	
5	When prompted for a movie id and rating to assign, choose “3” and “3”	Enter “3” twice	“”	“”	
6	When prompted for a movie id and rating to assign, choose “4” and “4”	Enter “4” twice	“”	“”	
7	When prompted for a movie id and rating to assign, choose “5” and “5”	Enter “5” twice	“”		
8	When prompted for a movie id and rating to assign, choose “1” and “5”	Enter “1” and then “5”	“”	“”	So far, we have only added new ratings so we will now begin updating existing ratings. Again, a screenshot of the data file is shown at the end for readability purposes
9	When prompted for a movie id and rating to assign, choose “2” and “4”	Enter “2” and then “4”	“”	“”	
10	When prompted for a movie id and rating to assign, choose “3” and “1”	Enter “3” and then “1”	“”	“”	
11	When prompted for a movie id	Enter “4” and then	“”	“”	

	and rating to assign, choose “4” and “2”	“2”			
12	When prompted for a movie id and rating to assign, choose “5” and “1”	Enter “5” and then “1”	“”		

Post condition(s) for Test: Restore my_ratings.txt

Test Stage: System Testing

Test Date: Dec 5, 2025

Test Case ID#: 9

Name of Tester: Ibrahim Awan

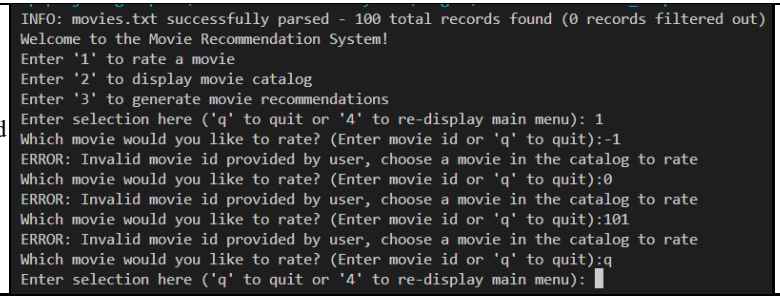
Test Description: Validation of REQ-10: The system must prevent users from rating a movie that does not exist in the master catalog. In other words, if the movie id provided by the user does not match any movie id present in movies.txt, the system should not modify my_ratings.txt. Instead, it should notify the user that they are attempting to rate an invalid movie and re-prompt the user for a movie id. The system must not crash if the user provides an invalid movie id.

Test Strategy: Attempt to rate movies with id as -1, 0, and 101 and verify appropriate error messages are displayed in the terminal

Results (P/F): P

Preconditions for Test: Use default movies.txt and my_ratings.txt files

Step #	Test Step Description	Test Data	Expected Result	Actual Result (terminal output)	Notes
1	Launch the application	Default data files	Normal startup	Normal startup	
2	When prompted for a selection, choose option “1”	Enter “1”	System prompts the user to enter the movie id to rate	Matches expectation	Because this is a required step if the user wants to repeatedly rate movies, it is only shown once here to avoid duplication
3	When prompted for a movie id, choose “-1”	Enter “-1”			Console output shown in the last row
4	When prompted for a movie id, choose “0”	Enter “0”			

5	When prompted for a movie id, choose “101”	Enter “101”	For all 3 cases, system displays appropriate messages and does not proceed with rating management		

Post condition(s) for Test: N/A

Test Stage: System Testing

Test Date: Dec 5, 2025

Test Case ID#: 10

Name of Tester: Ibrahim Awan

Test Description: Validation of REQ-11: The system must prevent users from assigning an invalid rating to a movie. The system must verify that the user-provided rating is an integer between 1 and 5. If this is not the case, the system should re-prompt the user to enter a valid rating (or exit to the main menu). If and only if the user enters a valid rating will the system allow writing to my_ratings.txt file.

Test Strategy: Attempt to rate movies with ratings as -1, 0, and 6 and verify appropriate error messages are displayed in the terminal

Results (P/F): P

Preconditions for Test: Use default movies.txt and my_ratings.txt files

Step #	Test Step Description	Test Data	Expected Result	Actual Result (terminal output)	Notes
1	Launch the application	Default data files	Normal startup	Normal startup	
2	When prompted for a selection, choose option “1”	Enter “1”	System prompts the user to enter the movie id to rate	Matches expectation	Because this is a required step if the user wants to repeatedly rate movies, it is only shown once here to avoid duplication
3	When prompted for a movie id, choose “5”	Enter “5”			This is also a required step since we’ll be using movie id 5 for this test case so it is only shown once here to avoid duplication
3	When prompted for a movie rating, choose “-1”	Enter “-1”			Console output shown in the last row
4	When prompted for a movie rating, choose “0”	Enter “0”			

5	When prompted for a movie rating, choose “6”	Enter “6”	For all 3 cases, system displays appropriate messages and does not proceed with rating management	INFO: movies.txt successfully parsed - 100 total records found (0 records filtered out) Welcome to the Movie Recommendation System! Enter '1' to rate a movie Enter '2' to display movie catalog Enter '3' to generate movie recommendations Enter selection here ('q' to quit or '4' to re-display main menu): 1 Which movie would you like to rate? (Enter movie id or 'q' to quit):5 How would you rate this movie? (Enter 1-5 (inclusive) for rating or 'q' to quit):-1 ERROR: Invalid movie rating provided by user, provide a rating between 1-5 (inclusive) Which movie would you like to rate? (Enter movie id or 'q' to quit):5 How would you rate this movie? (Enter 1-5 (inclusive) for rating or 'q' to quit):0 ERROR: Invalid movie rating provided by user, provide a rating between 1-5 (inclusive) Which movie would you like to rate? (Enter movie id or 'q' to quit):5 How would you rate this movie? (Enter 1-5 (inclusive) for rating or 'q' to quit):6 ERROR: Invalid movie rating provided by user, provide a rating between 1-5 (inclusive) Which movie would you like to rate? (Enter movie id or 'q' to quit):q	

Post condition(s) for Test: N/A

Test Stage: System Testing

Test Date: Dec 5, 2025

Test Case ID#: 11

Name of Tester: Ibrahim Awan

Test Description: Validation of REQ-12: The system must be able to generate movie recommendations using user ratings stored within my_ratings.txt. The algorithm exercised by the recommendation engine should assign similarity scores to each unwatched movie in movies.txt based on user ratings and the top 5 movies with the highest similarity scores will then be displayed to the user.

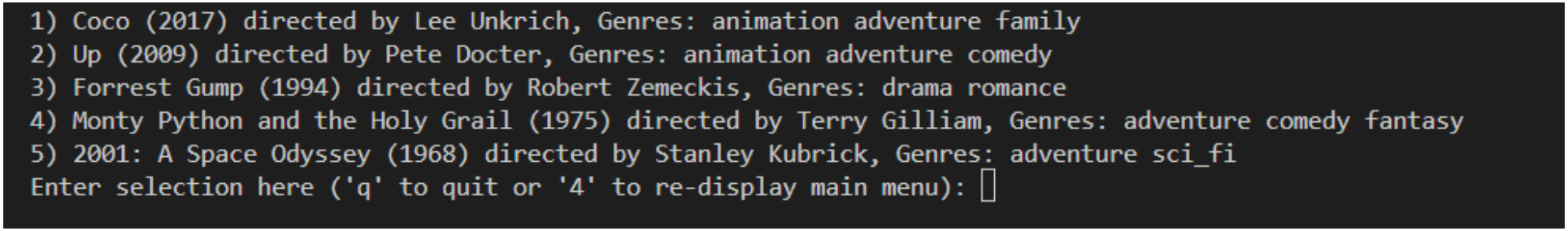
Test Strategy: Let's assume the user loves adventure and comedy films. Set up data in my_ratings.txt as shown and ensure the recommended movies are logical

Results (P/F): P

Preconditions for Test: Use default movies.txt and delete the default my_ratings.txt file to simulate cold-start

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Launch the application	Default movies.txt	On application startup, the terminal displays a message acknowledging a cold start. A my_ratings.txt file is created in the data/ folder	Matches expectation	
2	Insert custom ratings into my_ratings.txt to simulate user who loves adventure and comedy films	data > my_ratings.txt 1 movie_id,rating,timestamp 2 39,5,2025-12-06 3 45,5,2025-12-06 4 24,4,2025-12-06 5 42,4,2025-12-06 6 36,3,2025-12-06 7 44,3,2025-12-06	User is able to repeatedly select option "1" to rate movies	Matches expectation	Highly ranked films (text format): 39,Toy Story,John Lasseter,1995,Animation;Adventure;Comedy,4.5,1876 45,Toy Story 3,Lee Unkrich,2010,Animation;Adventure;Comedy,4.5,1654 24,Back to the Future,Robert Zemeckis,1985,Adventure;Comedy;Sci-Fi,4.6,2134 42,The Grand Budapest Hotel,Wes Anderson,2014,Adventure;Comedy;Crime,4.4,1234 36,Avengers: Infinity War,Anthony Russo,2018,Action;Adventure;Sci-Fi,4.4,3234 44,Eternal Sunshine of the Spotless Mind,Michel Gondry,2004,Drama;Romance;Sci-Fi,4.5,1123
3	When prompted for a	Enter "3"	The generated	Shown below	Coco was assigned the highest similarity score of 0.42 due to matching

	selection, choose option “3”		movie recommendations are geared towards adventure and comedy films		genres and director (Lee Unkrich) 2001: A Space Odyssey was assigned a similarity score of 0.21 due to matching genre (adventure) I was slightly confused why Forrest Gump was ranked at 3 but realized this is due to matching director (Robert Zemeckis) TLDR: Results make sense



Post condition(s) for Test: Restore default my_ratings.txt

Test Stage: System Testing

Test Date: Dec 5, 2025

Test Case ID#: 12

Name of Tester: Ibrahim Awan

Test Description: Validation of REQ-12: The system must be able to generate movie recommendations using user ratings stored within my_ratings.txt. The algorithm exercised by the recommendation engine should assign similarity scores to each unwatched movie in movies.txt based on user ratings and the top 5 movies with the highest similarity scores will then be displayed to the user.

Test Strategy: Let's assume the user loves crime and thriller films. Set up data in my_ratings.txt as shown and ensure the recommended movies are logical

Results (P/F): P

Preconditions for Test: Use default movies.txt and delete the default my_ratings.txt file to simulate cold-start

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Launch the application	Default movies.txt	On application startup, the terminal displays a message acknowledging a cold start. A my_ratings.txt file is created in the data/ folder	Matches expectation	
2	Insert custom ratings into my_ratings.txt to simulate user who loves crime and thriller films	data > my_ratings.txt movie_id,rating,timestamp 72,5,2025-12-06 48,5,2025-12-06 73,5,2025-12-06 3,5,2025-12-06 37,4,2025-12-06 35,4,2025-12-06 21,5,2025-12-06 18,4,2025-12-06	User is able to repeatedly select option "1" to rate movies	Matches expectation	Highly ranked films (text format): 72,No Country for Old Men,Joel Coen,2007,Crime;Drama;Thriller,4.4,1234 48,Reservoir Dogs,Quentin Tarantino,1992,Crime;Thriller,4.4,1234 73,Kill Bill: Vol. 1,Quentin Tarantino,2003,Action;Crime;Thriller,4.3,1543 3,The Dark Knight,Christopher Nolan,2008,Action;Crime;Drama,4.7,3421 37,The Dark Knight Rises,Christopher Nolan,2012,Action;Thriller,4.4,2654 35,Joker,Todd Phillips,2019,Crime;Drama;Thriller,4.3,2765 21,The Departed,Martin Scorsese,2006,Crime;Drama;Thriller,4.6,1654 18,Parasite,Bong Joon Ho,2019,Drama;Thriller,4.6,1876

3	When prompted for a selection, choose option “3”	Enter “3”	The generated movie recommendations are geared towards crime and thriller films	Shown below	Pulp fiction was assigned the highest similarity score of 0.51 due to matching genre and director (Quentin Tarantino) Taxi Driver was assigned a similarity score of 0.39 due to matching genre and director (Martin Scorsese) Christopher Nolan movies also made it in the top 5 since the user rated both Dark Knight movies highly TLDR: Results make sense

```
1) Pulp Fiction (1994) directed by Quentin Tarantino, Genres: crime drama
2) Memento (2000) directed by Christopher Nolan, Genres: mystery thriller
3) Fargo (1996) directed by Joel Coen, Genres: crime drama thriller
4) Inception (2010) directed by Christopher Nolan, Genres: action sci_fi thriller
5) Taxi Driver (1976) directed by Martin Scorsese, Genres: crime drama
Enter selection here ('q' to quit or '4' to re-display main menu):
```

Post condition(s) for Test: Restore default my_ratings.txt

Test Stage: System Testing

Test Date: Dec 5, 2025

Test Case ID#: 13

Name of Tester: Ibrahim Awan

Test Description: Validation of REQ-13: The system must be able to generate movie recommendations if there are no user ratings yet by considering the average rating and number of ratings of all movies within the master catalog.

Test Strategy: Run movie recommendation engine when my_ratings.txt is non-existent (i.e. it is created by the program, but the user does not rate any movies)

Results (P/F): P

Preconditions for Test: Use default movies.txt and delete the default my_ratings.txt file to simulate cold-start

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Launch the application	Default movies.txt	On application startup, the terminal displays a message acknowledging a cold start. A my_ratings.txt file is created in the data/ folder	Matches expectation	
2	When prompted for a selection, choose option “3”	Enter “3”	The generated movie recommendations are sorted by average rating and then by number of ratings	Shown below	The results make sense after consideration of avg_rating and num_ratings in the movies.txt file

```
Enter '1' to rate a movie
Enter '2' to display movie catalog
Enter '3' to generate movie recommendations
Enter selection here ('q' to quit or '4' to re-display main menu): 3
INFO: Generating movie recommendations...
1) The Shawshank Redemption (1994) directed by Frank Darabont, Genres: drama
2) The Lord of the Rings: The Return of the King (2003) directed by Peter Jackson, Genres: adventure drama fantasy
3) The Godfather (1972) directed by Francis Ford Coppola, Genres: crime drama
4) Schindler's List (1993) directed by Steven Spielberg, Genres: biography drama history
5) The Dark Knight (2008) directed by Christopher Nolan, Genres: action crime drama
Enter selection here ('q' to quit or '4' to re-display main menu):
```

Post condition(s) for Test: Restore default my_ratings.txt
